

TIANLIN LIU

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RESEARCH INTERESTS

Large Language Models, Reinforcement Learning, Text Diffusion

EMPLOYMENT

Nov 2024–now	Google DeepMind Research Scientist
Sept–Dec 2023	Google DeepMind Research Scientist Intern
July–Sept 2023	Google Brain Research Scientist Intern

EDUCATION

2020–2024	University of Basel , Switzerland Ph.D. Computer Science Advisor: Ivan Dokmanić
2017–2019	Jacobs University Bremen , Germany M.S. Data Engineering Advisor: Herbert Jaeger
2013–2016	Jacobs University Bremen , Germany B.S. Mathematics Advisor: Götz Pfander

AWARDS AND HONORS

2023, 2024	Google Conference Scholarship
2022	Young Researcher, Heidelberg Laureate Forum.
2022	Travel Award, SIAM Conference on Imaging Science.
2021	Oberwolfach Leibniz Graduate Student Grant, Oberwolfach Institute.
2019	Dean's Award for Outstanding Master's Thesis [4], Jacobs Univ. Bremen.
2019	IEEE NER 2019 Best Paper Finalist Award [2].
2018	SmartStart-1 Fellowship, Bernstein Network and Volkswagen Foundation.

SELECTED PUBLICATIONS

Full list: [Google Scholar](#)

(★ indicates equal contributions)

- [18] **Autoregressive Language Models are Secretly Energy-Based Models: Insights into the Lookahead Capabilities of Next-Token Prediction**
Mathieu Blondel, Michael E Sander, Germain Vivier-Ardisson, **Tianlin Liu**, Vincent Roulet.
In Proc. of the International Conference on Machine Learning (ICML), 2026.
- [17] **Joint learning of energy-based models and their partition function**
Michael E Sander, Vincent Roulet, **Tianlin Liu**, Mathieu Blondel.
In Proc. of the International Conference on Machine Learning (ICML), 2025.
- [16] **Loss Functions and Operators Generated by f-Divergences**
Vincent Roulet, **Tianlin Liu**, Nino Vieillard, Michael E Sander, Mathieu Blondel.
In Proc. of the International Conference on Machine Learning (ICML), 2025.
- [15] **Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise**
Enea Monzio Compagnoni, **Tianlin Liu**, Rustem Islamov, Frank Norbert Proske, Antonio Orvieto, Aurelien Lucchi.
In Proc. of the International Conference on Learning Representations (ICLR), 2025.
- [14] **LLMs assist NLP researchers: Critique paper (meta-) reviewing.**
Jiangshu Du, ..., Qin Liu, **Tianlin Liu**, ..., Wenpeng Yin.
In Proc. of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.
- [13] **Decoding-time Realignment of Language Models.**
Tianlin Liu, Shangmin Guo, Leonardo Bianco, Daniele Calandriello, Quentin Berthet, Felipe Llinares, Jessica Hoffmann, Lucas Dixon, Michal Valko, Mathieu Blondel.
In Proc. of the International Conference on Machine Learning (ICML), 2024.
Spotlight presentation
- [12] **Routers in Vision Mixture of Experts: An Empirical Study.**
Tianlin Liu, Mathieu Blondel, Carlos Riquelme, Joan Puigcerver.
Transactions on Machine Learning Research (TMLR), 2024.
- [11] **The N Implementation Details of RLHF with PPO.**
Shengyi Huang, **Tianlin Liu**, Leandro Von Werra.
The Blogpost Track of the International Conference on Learning Representations (ICLR), 2024.
Spotlight presentation
- [10] **WaveBench: Benchmarking Data-driven Solvers for Linear Wave Propagation PDEs.**
Tianlin Liu[★], Jose Antonio Lara Benitez[★], Florian Faucher[★], AmirEhsan Khorashadizadeh, Maarten V. de Hoop, Ivan Dokmanić.
Transactions on Machine Learning Research (TMLR), 2024.
- [9] **Sparsity-Constrained Optimal Transport.**
Tianlin Liu, Joan Puigcerver, and Mathieu Blondel.
In Proc. of the International Conference on Learning Representations (ICLR), 2023.
Spotlight presentation
- [8] **Learning Multiscale Convolutional Dictionaries for Image Reconstruction.**
Tianlin Liu, Anadi Chaman, David Belius, and Ivan Dokmanić.
IEEE Transactions on Computational Imaging, 2022.
- [7] **Universal Approximation Under Constraints is Possible with Transformers.**
Anastasis Kratsios, Behnoosh Zamanlooy, **Tianlin Liu**, and Ivan Dokmanić.
In Proc. of the International Conference on Learning Representations (ICLR), 2022.
Spotlight presentation

- [6] **Finding Trainable Sparse Networks through Neural Tangent Transfer.**
Tianlin Liu and Friedemann Zenke.
 In Proc. of the International Conference on Machine Learning (ICML), 2020.
- [5] **Causally Denoise Word Embeddings Using Half-Sibling Regression.**
 Zekun Yang[★] and **Tianlin Liu[★]**.
 In Proc. of the AAAI Conference on Artificial Intelligence (AAAI), 2020.
- [4] **Harnessing Slow Dynamics in Neuromorphic Computation.**
Tianlin Liu. Master's thesis, 2019.
Dean's Prize for Outstanding Master's Thesis
- [3] **Continual Learning for Sentence Representations Using Conceptors.**
Tianlin Liu, Lyle Ungar, and João Sedoc.
 In Proc. of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2019.
- [2] **Reservoir Transfer on Analog Neuromorphic Hardware.**
 Xu He, **Tianlin Liu**, Fatemeh Hadaeghi, and Herbert Jaeger.
 In Proc. of the International IEEE Conference on Neural Engineering (NER), 2019.
Best Paper Finalist Award (3rd place out of 467 accepted papers)
- [1] **Unsupervised Post-processing of Word Vectors via Conceptor Negation.**
Tianlin Liu, Lyle Ungar, and João Sedoc.
 In Proc. of the AAAI Conference on Artificial Intelligence (AAAI), 2019.

OPEN-SOURCE CONTRIBUTION

[huggingface/transformers](#): transformers library
[huggingface/trl](#): lightweight library for RLHF
[PythonOT/POT](#): python optimal transport
[google-research/vmoe](#): vision mixture of experts models
[google/jaxopt](#): differentiable optimizers in JAX

JOURNAL AND CONFERENCE REVIEWING

CONFERENCE	ICML (2021–2025), NeurIPS (2021–2023), ICLR (2022, 2023), LoG (2022)
JOURNAL	Journal of Machine Learning Research, IEEE Transactions on Signal Processing, IEEE Signal Processing Letters, IEEE Open Journal of Signal Processing